Experiment 1

Ex. No.: 1

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

To study the purpose of digital forensics and understand the main commercial, open-source, and field tools used during an investigation.

Tools required

1. Commercial suites: EnCase Forensic, Forensic Toolkit (FTK), and ProDiscover.
2. Open-source platforms: The Sleuth Kit, Helix, and Knoppix.
3. Field utilities: FTK Imager, Log Parser Lizard, Noxcivis Field Toolkit, Active@ Partition Recovery, and Autopsy.

Idea/Theory

1. Digital forensics applies repeatable methods to identify, collect, examine, analyse, and present digital evidence.
2. Useful traces may remain in file systems, logs, registry data, installed software, malware artefacts, and deleted-file remnants.
3. A sound investigation preserves the original evidence, records the chain of custody, and verifies copies with cryptographic hashes.
4. Commercial suites combine acquisition, examination, reporting, and training support; open tools expose many of the same underlying techniques.

Experiment 1 — continued

Procedure/Steps

1. Define the legal authority, case scope, and evidence sources before collection.
2. Identify relevant computers, storage media, accounts, logs, and removable devices.
3. Create a forensic image or another approved acquisition; keep the original evidence protected and work from a verified copy.
4. Examine file systems, metadata, logs, deleted items, and application artefacts with suitable tools.
5. Correlate findings, document each action, and prepare a clear report that separates facts from interpretation.

Observation/Output

The tools named in the manual can be grouped by role as follows:

Category	Tool	Purpose in an investigation
Commercial suite	EnCase Forensic	Acquires, searches, examines, and reports digital evidence through an integrated forensic workflow.
Commercial suite	Forensic Toolkit (FTK)	Indexes evidence, supports file and artefact analysis, and provides reporting; FTK Imager is its acquisition utility.
Commercial suite	ProDiscover	Supports disk imaging, file-system examination, search, and case reporting.
Open source	The Sleuth Kit	Provides command-line libraries and tools for examining disk images and file systems.
Open source	Helix	A forensics-focused environment historically used for live response and incident investigation.
Open source	Knoppix	A bootable Linux environment that can host recovery and forensic utilities.

Experiment 1 — continued

Observation/Output (continued)

Category	Tool	Purpose in an investigation
Field tool	FTK Imager	Creates forensic images, previews evidence, and calculates verification hashes.
Field tool	Log Parser Lizard	Provides a graphical interface for querying and reviewing log data.
Field tool	Noxcivis Field Toolkit	Supports field collection of information from a computer through an examiner-oriented interface.
Field tool	Active@ Partition Recovery	Locates and recovers deleted or damaged partitions.
Field tool	Autopsy	Graphical digital-forensics platform built around The Sleuth Kit and other modules; supports disk, file, and media-card investigations.

Result

The study of digital forensics concepts and the listed commercial, open-source, and field investigation tools was completed.